NUTM

NIGERIAN UNIVERSITY OF TECHNOLOGY & MANAGEMENT

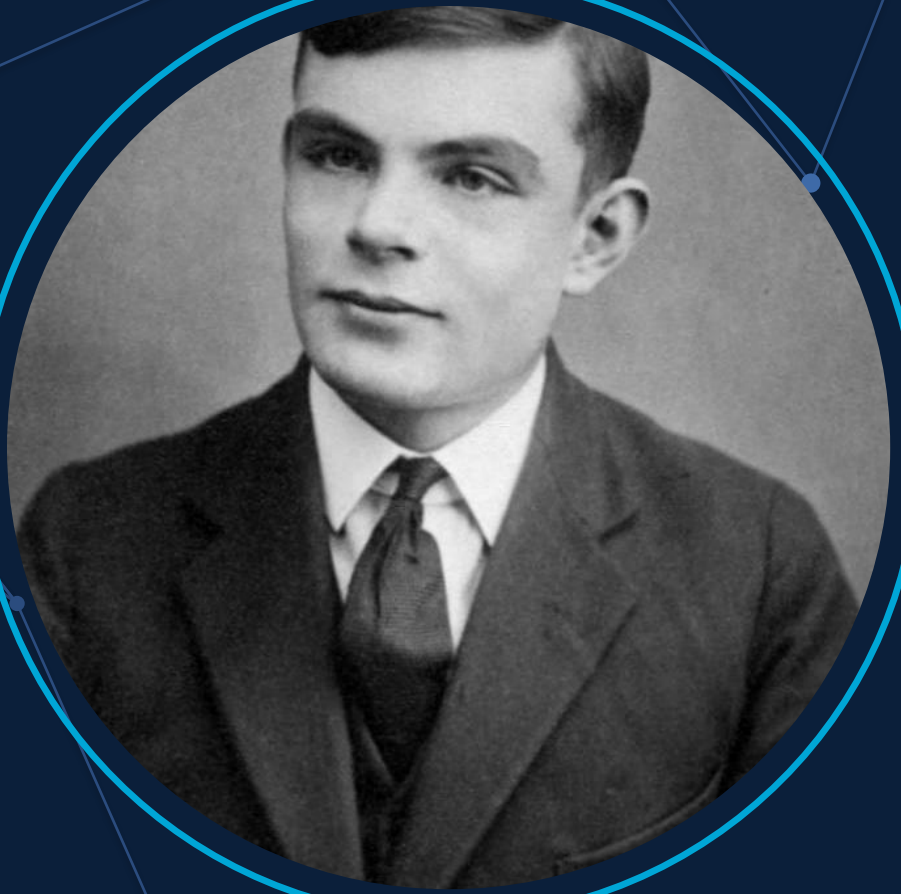



# Artificial Intelligence

By

Charles Igah

# Foundations of Artificial Intelligence



# Let's discuss



## Is your smartphone intelligent? What does it do that seems ‘smart’ — and what can't it do?

**THINK**

List three ‘smart’ things your phone does.

**PAIR**

Which of them learn from you over time?

**SHARE**

What would convince you that a machine is really intelligent?

# Data



It’s all about the
Data



# Data All Around

- Lots of data is being collected and warehoused
  - Web data, e-commerce
  - Financial transactions, bank/credit transactions
  - Online trading and purchasing
  - Social Network

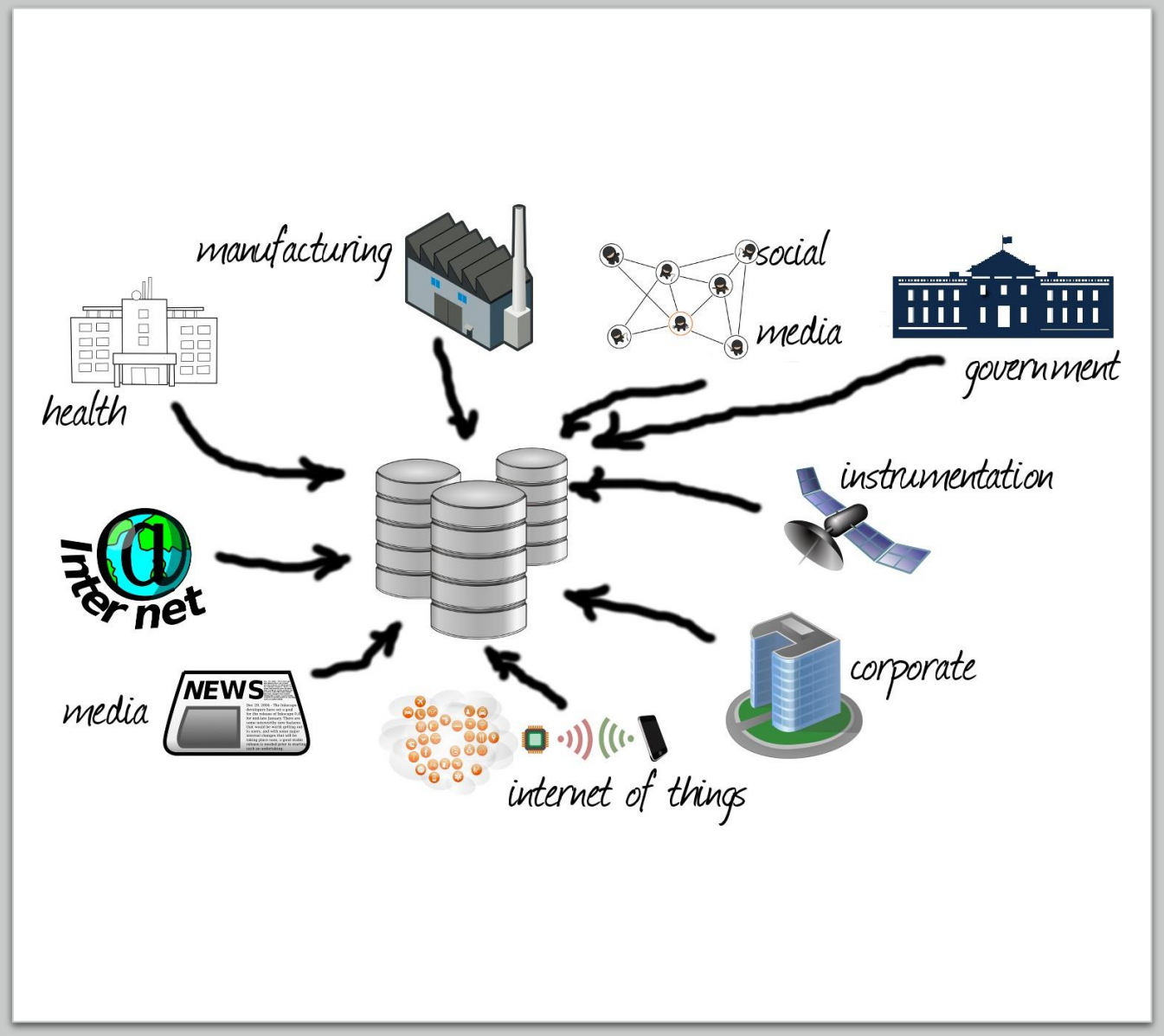









# Data can be got from many sources.



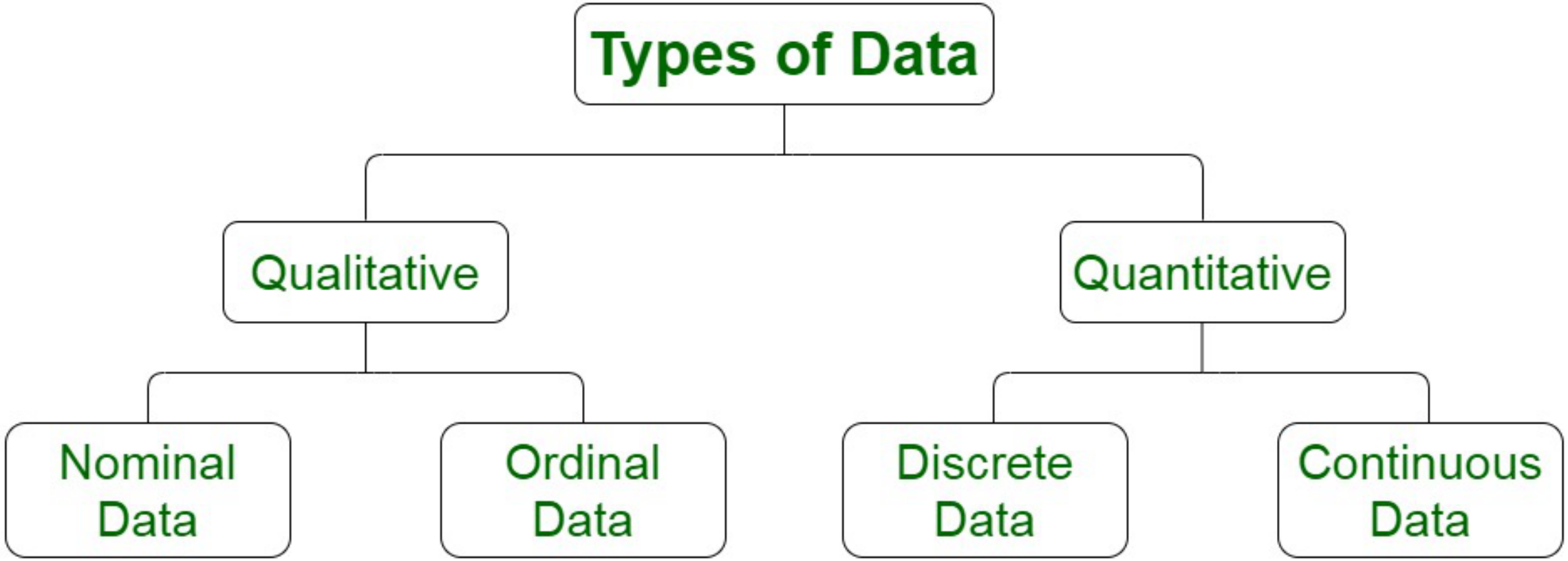
# Types of Data



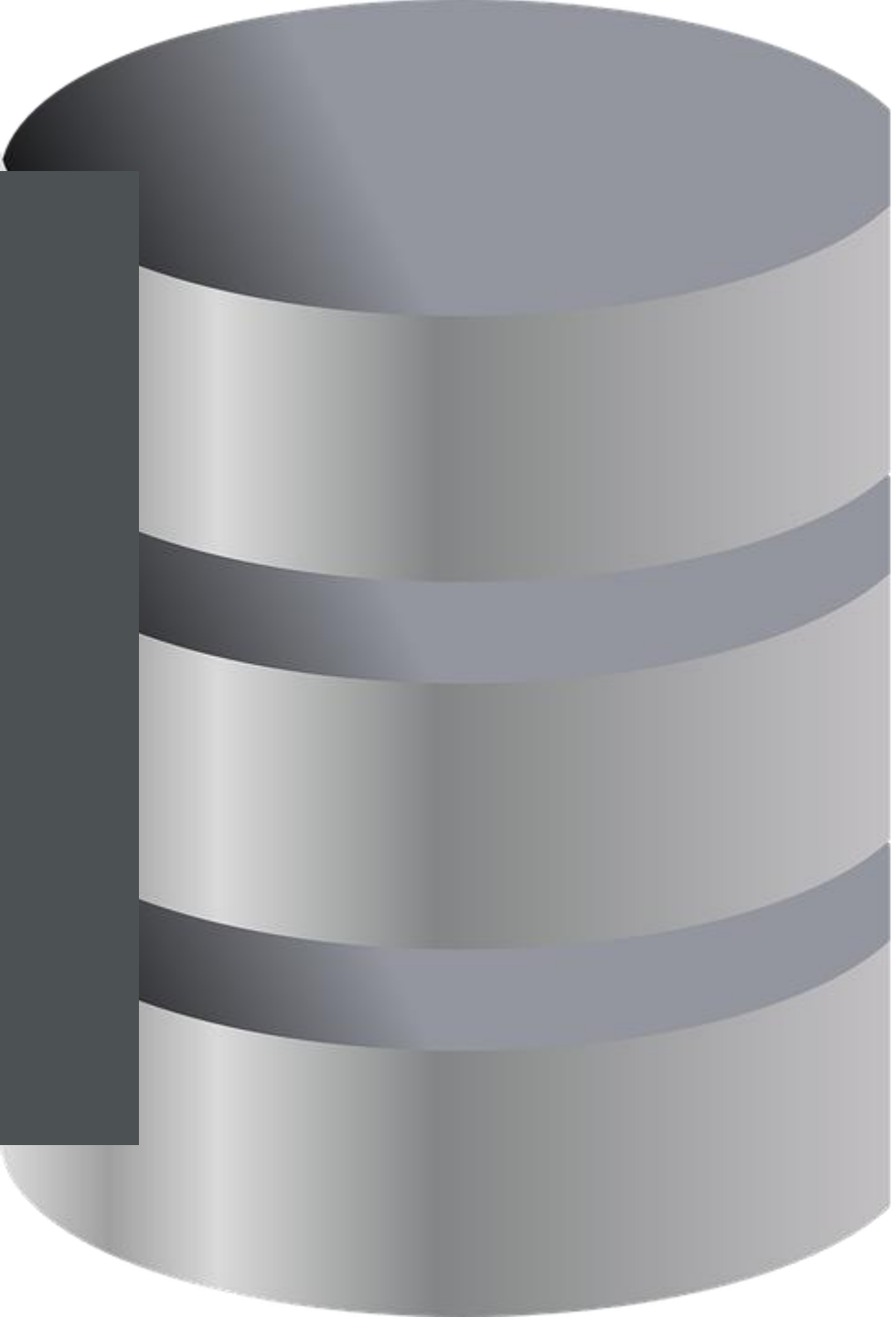
Qualitative: …Descriptive data which can be observed but not measured.



Quantitative: …Numerical data which can be measured.


Types of Data
Qualitative
Quantitative
Nominal Data
Ordinal Data
Discrete Data
Continuous Data

| Types of Data | Category | Examples |
| --- | --- | --- |
| **Qualitative Data** | **Nominal Data** | Gender (Women, Men), Hair color (Blonde, Brown), Ethnicity (Hispanic, Asian) |
| | **Ordinal Data** | First, second, and third; Letter grades (A, B, C); Economic status (low, medium) |
| **Quantitative Data** | **Discrete Data** | The number of students in a class, The number of workers in a company, The number of cars |
| | **Continuous Data** | The height of children, weight, temperature. |



# Data Format

STRUCTURED DATA

UNSTRUCTURED DATA

SEMI-STRUCTURED DATA

# Data Format



**Structured Data:**

Data that is organized in a fixed format, often in tables.

Examples: Databases, spreadsheets.



**Unstructured Data:**

Data that does not have a predefined format.

Examples: Texts, social media posts, videos, images.

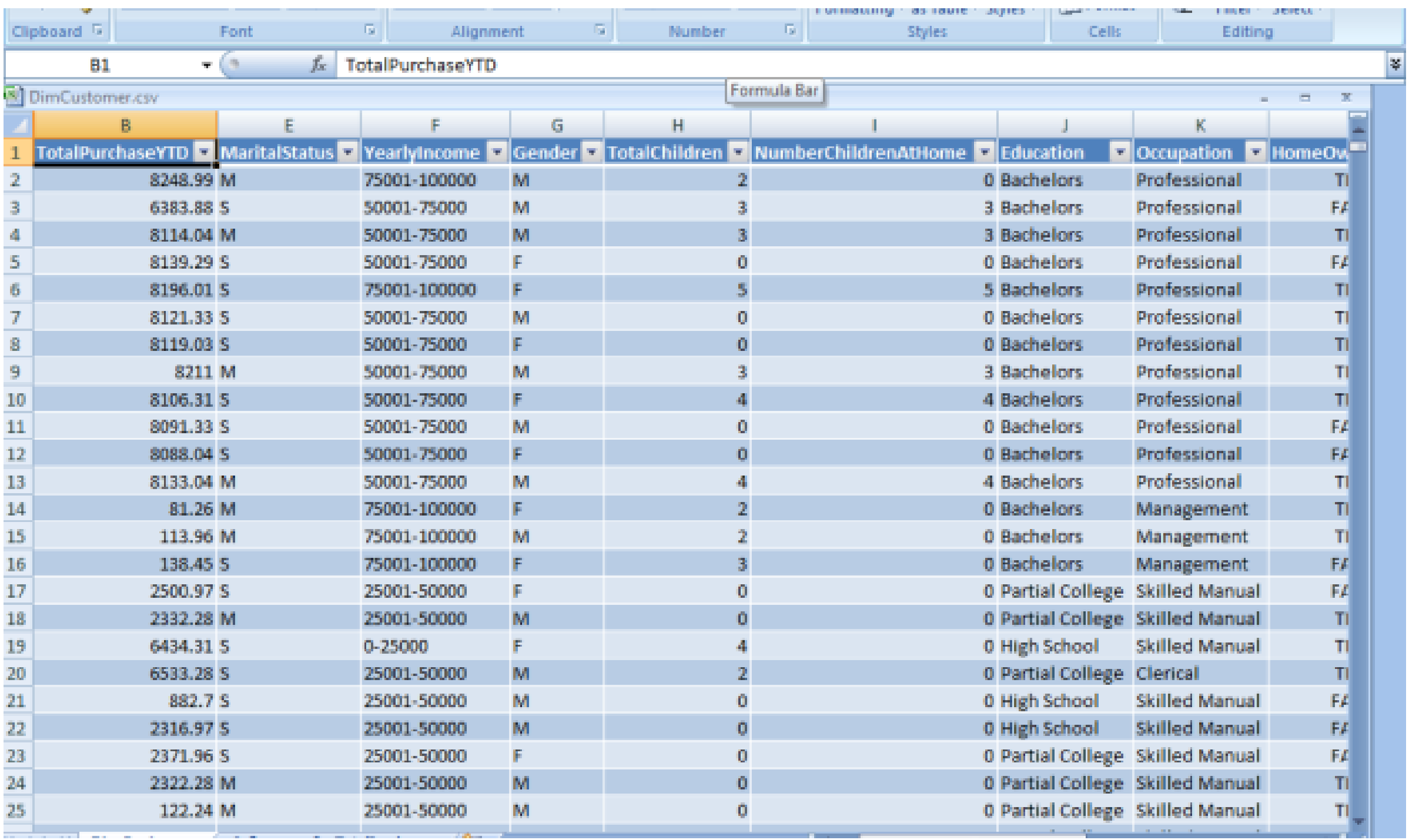


**Semi-Structured Data:**

Data that does not conform to a rigid structure but has some organizational properties.

Examples: JSON, XML files.


<XML>
JSON
Structured Data
Textual Data
Image File
Video
Audio
XML Data
JSON Data
Sensor Data




| | B | E | F | G | H | I | J | K | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | TotalPurchaseYTD | MaritalStatus | YearlyIncome | Gender | TotalChildren | NumberChildrenAtHome | Education | Occupation | HomeOv |
| 2 | 8248.99 | M | 75001-100000 | M | 2 | 0 | Bachelors | Professional | Tl |
| 3 | 6383.88 | S | 50001-75000 | M | 3 | 3 | Bachelors | Professional | FA |
| 4 | 8114.04 | M | 50001-75000 | M | 3 | 3 | Bachelors | Professional | Tl |
| 5 | 8139.29 | S | 50001-75000 | F | 0 | 0 | Bachelors | Professional | FA |
| 6 | 8196.01 | S | 75001-100000 | F | 5 | 5 | Bachelors | Professional | Tl |
| 7 | 8121.33 | S | 50001-75000 | M | 0 | 0 | Bachelors | Professional | Tl |
| 8 | 8119.03 | S | 50001-75000 | F | 0 | 0 | Bachelors | Professional | Tl |
| 9 | 8211 | M | 50001-75000 | M | 3 | 3 | Bachelors | Professional | Tl |
| 10 | 8106.31 | S | 50001-75000 | F | 4 | 4 | Bachelors | Professional | Tl |
| 11 | 8091.33 | S | 50001-75000 | M | 0 | 0 | Bachelors | Professional | FA |
| 12 | 8088.04 | S | 50001-75000 | F | 0 | 0 | Bachelors | Professional | FA |
| 13 | 8133.04 | M | 50001-75000 | M | 4 | 4 | Bachelors | Professional | Tl |
| 14 | 81.26 | M | 75001-100000 | F | 2 | 0 | Bachelors | Management | Tl |
| 15 | 113.96 | M | 75001-100000 | M | 2 | 0 | Bachelors | Management | Tl |
| 16 | 138.45 | S | 75001-100000 | F | 3 | 0 | Bachelors | Management | FA |
| 17 | 2500.97 | S | 25001-50000 | F | 0 | 0 | Partial College | Skilled Manual | FA |
| 18 | 2332.28 | M | 25001-50000 | M | 0 | 0 | Partial College | Skilled Manual | Tl |
| 19 | 6434.31 | S | 0-25000 | F | 4 | 0 | High School | Skilled Manual | Tl |
| 20 | 6533.28 | S | 25001-50000 | M | 2 | 0 | Partial College | Clerical | Tl |
| 21 | 882.7 | S | 25001-50000 | M | 0 | 0 | High School | Skilled Manual | FA |
| 22 | 2316.97 | S | 25001-50000 | M | 0 | 0 | High School | Skilled Manual | FA |
| 23 | 2371.96 | S | 25001-50000 | F | 0 | 0 | Partial College | Skilled Manual | FA |
| 24 | 2322.28 | M | 25001-50000 | M | 0 | 0 | Partial College | Skilled Manual | Tl |
| 25 | 122.24 | M | 25001-50000 | M | 0 | 0 | Partial College | Skilled Manual | Tl |

```
scription,Director,Actors,Year,Runtime (Minutes),Rating,Votes,Revenue (Millions),Metascore
Galaxy,"Action,Adventure,Sci-Fi",A group of intergalactic criminals are forced to work together to stop a fanatical warrior from taking contro
ture,Mystery,Sci-Fi","Following clues to the origin of mankind, a team finds a structure on a distant moon, but they soon realize they are not
iller",Three girls are kidnapped by a man with a diagnosed 23 distinct personalities. They must try to escape before the apparent emergence of
omedy,Family","In a city of humanoid animals, a hustling theater impresario's attempt to save his theater with a singing competition becomes g
tion,Adventure,Fantasy",A secret government agency recruits some of the most dangerous incarcerated super-villains to form a defensive task fo
ction,Adventure,Fantasy",European mercenaries searching for black powder become embroiled in the defense of the Great Wall of China against a
y,Drama,Music",A jazz pianist falls for an aspiring actress in Los Angeles.,Damien Chazelle,"Ryan Gosling, Emma Stone, Rosemarie DeWitt, J.K.
A has-been actor best known for playing the title character in the 1980s detective series ""Mindhorn"" must work with the police when a serial
Z,"Action,Adventure,Biography","A true-life drama, centering on British explorer Col. Percival Fawcett, who disappeared while searching for a
nture,Drama,Romance","A spacecraft traveling to a distant colony planet and transporting thousands of people has a malfunction in its sleep ch
 and Where to Find Them,"Adventure,Family,Fantasy",The adventures of writer Newt Scamander in New York's secret community of witches and wizar
Biography,Drama,History",The story of a team of female African-American mathematicians who served a vital role in NASA during the early years
n,Adventure,Sci-Fi","The Rebel Alliance makes a risky move to steal the plans for the Death Star, setting up the epic saga to follow.",Gareth
,Adventure,Comedy","In Ancient Polynesia, when a terrible curse incurred by the Demigod Maui reaches an impetuous Chieftain's daughter's islar
,Comedy,Drama","Gloria is an out-of-work party girl forced to leave her life in New York City, and move back home. When reports surface that a
of Pets,"Animation,Adventure,Comedy","The quiet life of a terrier named Max is upended when his owner takes in Duke, a stray whom Max instantl
iography,Drama,History","WWII American Army Medic Desmond T. Doss, who served during the Battle of Okinawa, refuses to kill people, and become
tion,Thriller",The CIA's most dangerous former operative is drawn out of hiding to uncover more explosive truths about his past.,Paul Greengra
Drama","A five-year-old Indian boy gets lost on the streets of Calcutta, thousands of kilometers from home. He survives many challenges before
ystery,Sci-Fi","When twelve mysterious spacecraft appear around the world, linguistics professor Louise Banks is tasked with interpreting the
Drama,Thriller","Kenny Wells, a prospector desperate for a lucky break, teams up with a similarly eager geologist and sets off on a journey to
e Sea,Drama,A depressed uncle is asked to take care of his teenage nephew after the boy's father dies.,Kenneth Lonergan,"Casey Affleck, Michel
Crime,Drama,Horror","A cold-blooded predatory couple while cruising the streets in search of their next victim, will stumble upon a 17-year-ol
n,Adventure,Comedy","After the Bergens invade Troll Village, Poppy, the happiest Troll ever born, and the curmudgeonly Branch set off on a jou
: Resurgence,"Action,Adventure,Sci-Fi","Two decades after the first Independence Day invasion, Earth is faced with a new extra-Solar threat. E
Comedy,"Fiona visits Paris for the first time to assist her myopic Aunt Martha. Catastrophes ensue, mainly involving Dom, a homeless man who h
ginning,"Action,Adventure,Drama","In ancient India, an adventurous and daring man becomes involved in a decades old feud between two warring p
or,Thriller",A young woman must save herself and her friends from an ancient evil that stalks its victims through the real-life phenomenon of
"When three overworked and under-appreciated moms are pushed beyond their limits, they ditch their conventional responsibilities for a jolt of
,"Action,Adventure,Drama","When Callum Lynch explores the memories of his ancestor Aguilar and gains the skills of a Master Assassin, he disco
A holiday gathering threatens to go off the rails when Ned Fleming realizes that his daughter's Silicon Valley millionaire boyfriend is about
s,"Drama,Thriller","A wealthy art gallery owner is haunted by her ex-husband's novel, a violent thriller she interprets as a symbolic revenge
```

From r  Thu Oct 31 08:11:39 2002
Return-Path: <bensul2004nng@spinfinder.com>
X-Sieve: cmu-sieve 2.0
Return-Path: <bensul2004nng@spinfinder.com>
Message-Id: <200210311310.g9VDANt24674@bloodwork.mr.itd.UM>
From: "Mr. Ben Suleman" <bensul2004nng@spinfinder.com>
Date: Thu, 31 Oct 2002 05:10:00
To: R@M
Subject: URGENT ASSISTANCE /RELATIONSHIP (P)
MIME-Version: 1.0
Content-Type: text/plain;charset="iso-8859-1"
Content-Transfer-Encoding: 7bit
Status: O

Dear Friend,

I am Mr. Ben Suleman a custom officer and work as Assistant controller of the Customs and Excise department Of the Federal Ministry of Internal Affairs stationed at the

After the sudden death of the former Head of state of Nigeria General Sanni Abacha on June 8th 1998 his aides and immediate members of his family were arrested while tr

I declared only (5) five boxes to the government and withheld one (1) in my custody containing the sum of (US$30,000,000.00) Thirty Million United States Dollars Only,
to US$55 Billion Dollars (ref:ngrguardiannews.com) of July 2nd 1999. Even the London times of July 1998 reported that General Abacha has over US$3.Billion dollars in or
content will not be discovered. Now that all is calm, we (myself and two of my colleagues in the operations team) are now ready to move this box out of the country thro

However as government officials the Civil Service Code of Conduct does not allow us by law to operate any foreign account or own foreign investment and the amount of mo
cannot be more than our salary on the average, thus our handicapp and our need for your assistance to help collect and keep safely in your account this money.

Therefore we want you to assist us in moving this money out of Nigeria. We shall definitely compensate you handsomely for the assistance. We can do this by instructing
move the consignment to their affiliate branch office outside Nigeria through diplomatic means and the consignment will be termed as Precious Stones and Jewelleries" wh

This business is 100% risk free for you so please treat this matter with utmost confidentiality .If you indicate your interest to assist us please just e-mail me for mo

Expecting your response urgently.

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                <pose>Unspecified</pose>
                <truncated>0</truncated>
                <difficult>0</difficult>
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  "responses": [
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… back to AI

# A

# What is artificial intelligence?



# Key term



## Artificial intelligence (AI)

*noun · term coined by John McCarthy, 1955*

The science and engineering of making machines do tasks that would need **intelligence** if people did them: reasoning, learning, perceiving, understanding language and solving problems.

Google Maps predicting the fastest route through Lagos traffic, a bank flagging an unusual transfer, your phone recognising your face, a chatbot answering questions.

# Some well-known definitions of AI

| Who | Definition (paraphrased) | Emphasis |
|---|---|---|
| **John McCarthy (1955)** | The science and engineering of making intelligent machines | Engineering |
| **Marvin Minsky (1968)** | Making machines do things that would require intelligence if done by people | Human comparison |
| **Elaine Rich (1983)** | The study of how to make computers do things at which, at the moment, people are better | Moving target |
| **Russell & Norvig** | The study of agents that receive percepts and perform actions to achieve goals | Rational agents |
| **Nils Nilsson** | Activity devoted to making machines intelligent — able to function appropriately in their environment | Behaviour |

# What does ‘intelligence’ involve?



## Reasoning

Drawing conclusions from facts and rules.



## Learning

Improving from experience and examples.



## Perception

Making sense of images, sounds and sensors.



## Language

Understanding and producing speech and text.



## Problem solving

Finding a sequence of steps to reach a goal.



## Acting

Moving and manipulating objects (robotics).

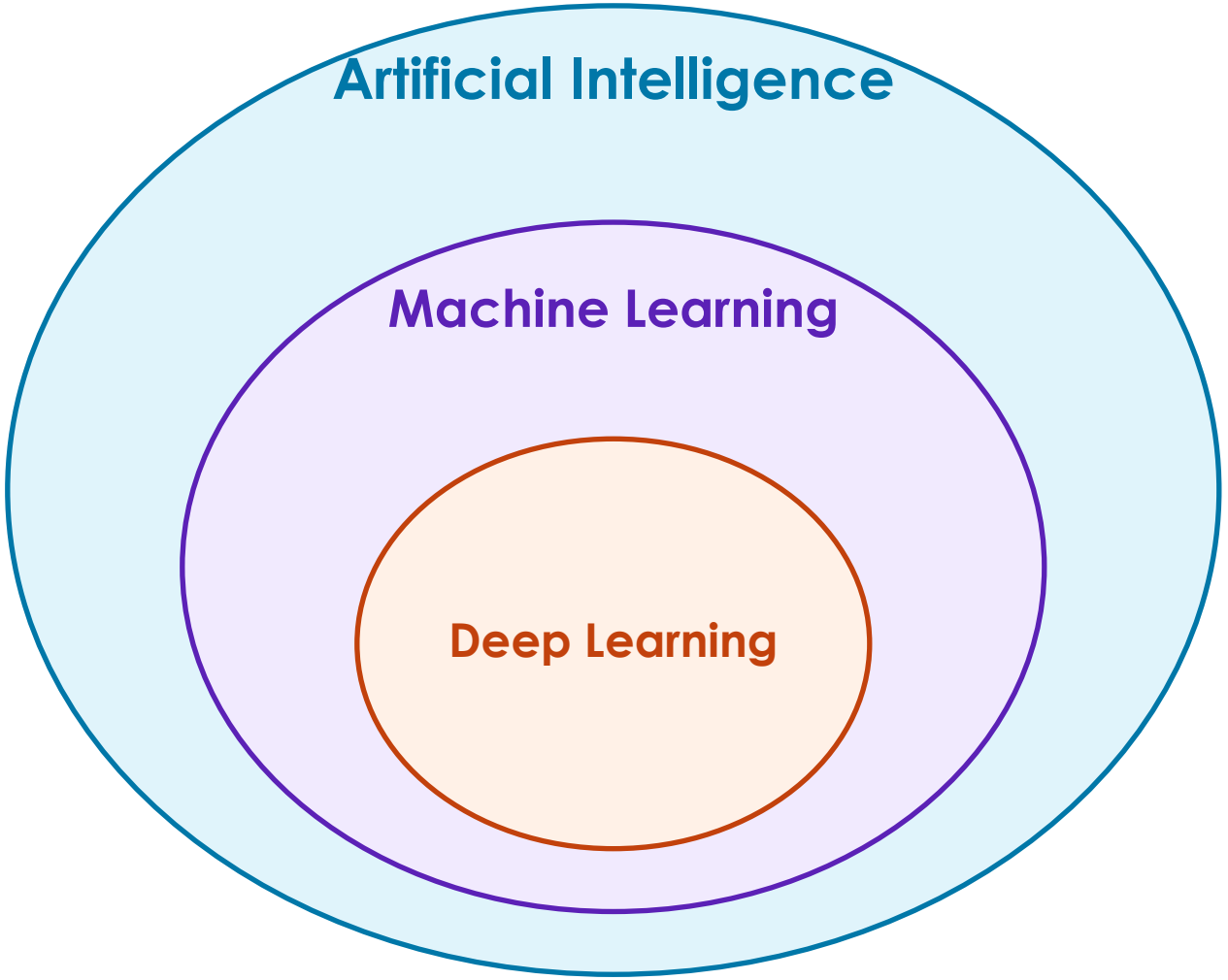
# Four ways to define AI (Russell & Norvig)

| | Like humans | Rationally (the ‘right’ thing) |
|---|---|---|
| THINKING | **Thinking humanly**<br>Model how people think: cognitive science. ‘Get inside the human mind.’ | **Thinking rationally**<br>Laws of thought: logic, correct inference from facts (Module 4). |
| ACTING | **Acting humanly**<br>Behave so well people can't tell it is a machine: the Turing test. | **Acting rationally**<br>Rational agents that act to achieve the best expected outcome (Module 2).<br>THIS COURSE'S FOCUS |

# Intelligent behaviour: what would count?

| Behaviour | Needs… | Can AI do it today? |
|---|---|---|
| **Recognise a friend's face in a crowd** | Perception | Yes — very well in good conditions |
| **Hold a conversation about the news** | Language, knowledge | Yes — large language models |
| **Fold laundry in any home** | Perception, dexterity, common sense | Only in limited lab settings |
| **Explain a joke in Pidgin** | Language, culture, common sense | Sometimes — often misses context |
| **Plan a cheap route Lagos → Abuja** | Search, planning | Yes |
| **Know when it doesn't know** | Self-assessment | Poorly — a major research problem |

# AI, machine learning and deep learning




## Artificial Intelligence (1956 →)

Any technique that makes machines act intelligently — including search, logic and hand-written rules.

## Machine Learning (1959 →)

Systems that learn patterns from data instead of following only hand-written rules.

## Deep Learning (2006/2012 →)

Machine learning with many-layered neural networks: vision, speech, translation, chatbots.

# Traditional programming vs machine learning

## Traditional programming

*Rules → answers*

- A programmer writes the **rules**
- Data + rules → **answers**
- e.g. ‘if score ≥ 70 then grade A’
- Great when rules are clear and stable

VS

## Machine learning

*Examples → rules*

- The computer **learns the rules** from examples
- Data + answers → **rules (a model)**
- e.g. learn spam from labelled emails
- Great when rules are fuzzy or change

# Why AI works so well now: three ingredients




## Data

Billions of photos, texts, transactions and sensor readings.



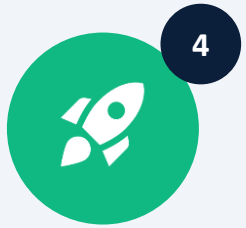

## Computing power

GPUs and cloud computers, millions of times faster than 1990s PCs.




## Algorithms

Better methods: deep learning, transformers, reinforcement learning.




## Result

AI that sees, translates and converses at near-human level in narrow tasks.

# AI you used this week (probably)

### Navigation

Google Maps predicts traffic and re-routes you around Lagos go-slows.

### Spam filters

Email providers learn what junk mail looks like.

### Face unlock

Phones recognise your face from the camera.

### Predictive text

Your keyboard guesses the next word.

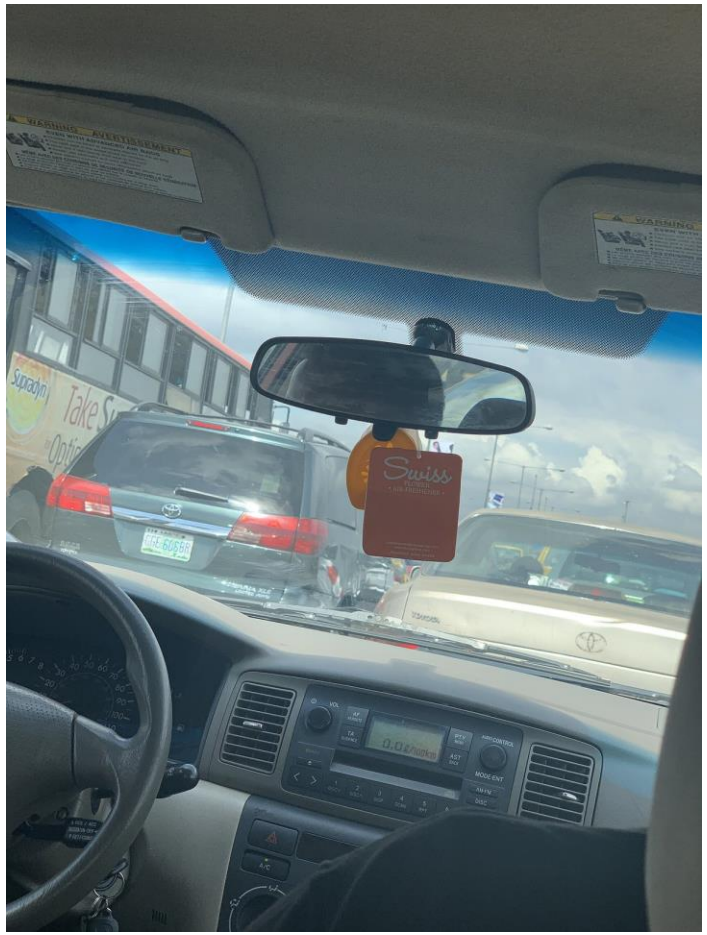
### Translation

Google Translate handles Hausa, Igbo and Yoruba.

### Recommendations

YouTube, TikTok and Netflix pick what to show you.

# AI at work on Lagos roads




*Traffic congestion in Lagos*

- Navigation apps combine **GPS data** from millions of phones to estimate traffic speed.
- They **predict** journey times and suggest faster routes.
- Underneath: **search algorithms** (Module 3) and **machine learning**.
- Ride-hailing apps also use AI to match drivers and set prices.

# See AI guess what you mean: AutoDraw

Follow these steps:

1. Open **AutoDraw** (Google) on your phone or laptop.
2. Scribble a rough object — a cup, a house, a bicycle.
3. Watch the AI **suggest** clean drawings of what it thinks you meant.
4. Which capability of intelligence is this: reasoning, perception, language or learning?

**Scan or click**



autodraw.com

**Open the link**

# Humans and machines: different strengths

## People are better at…

- **Common sense** and understanding context
- Learning from **a few** examples
- Creativity, empathy and moral judgement
- Coping with **new, messy** situations

VS

## Machines are better at…

- **Speed** and tireless repetition
- Remembering and searching **huge** amounts of data
- Exact calculation and consistency
- Spotting patterns in **millions** of records

CLASS ACTIVITY

# AI or not AI?



**10 min**

**Pairs**

**Outcome:** See that ‘AI’ is a spectrum, not a yes/no label.

1. In pairs, sort these into **AI** and **not AI**: a calculator, Google Maps, a washing-machine timer, a face filter on Snapchat, an ATM, a chess app, Spotify recommendations, a thermostat.
2. For each AI example, name the **capability** it uses (learning, perception, language, reasoning…).
3. Find one example where your pair **disagrees** — why?
4. Share your hardest case with the class.

B

# A history of AI

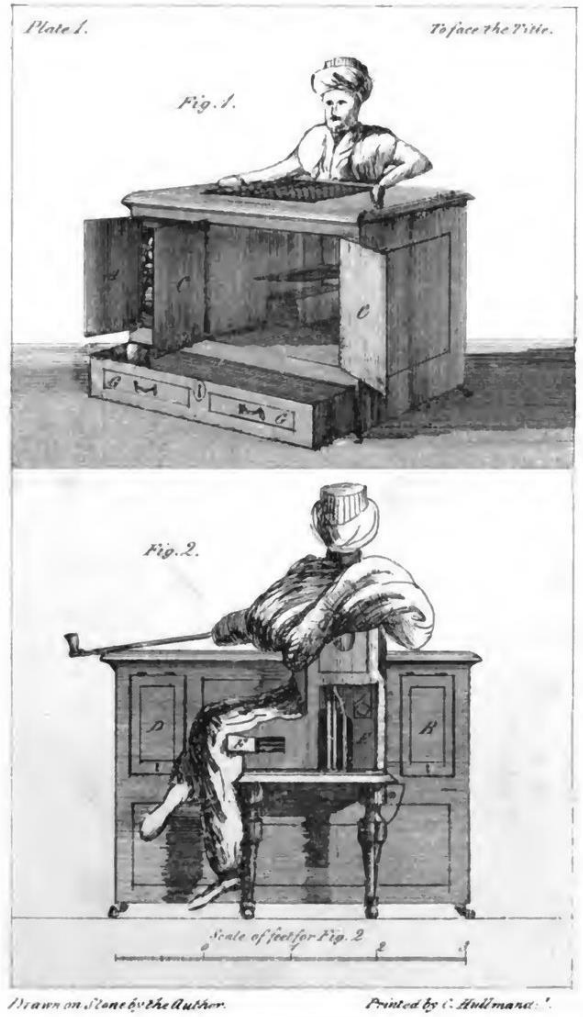
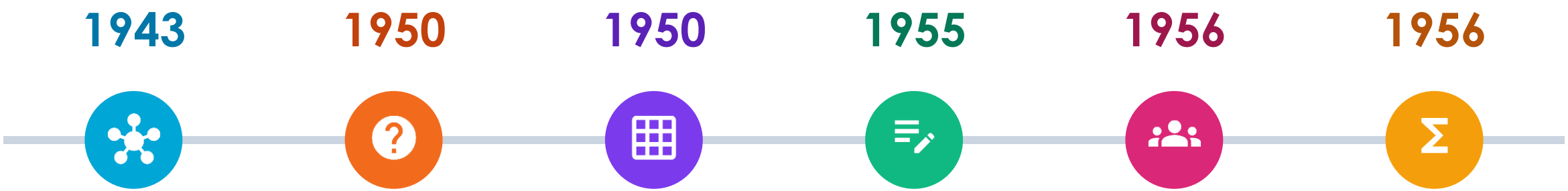



*Image: Willis, Robert, 1800-1875 · Public domain · Wikimedia Commons*

Digitized by Google

# 1943–1956: the birth of AI



**1943**

**Artificial neuron**

McCulloch & Pitts model a neuron in logic

**1950**

**Turing's question**

‘Can machines think?’ — the imitation game

**1950**

**Chess by computer**

Shannon shows how a program could play chess

**1955**

**The term ‘AI’**

McCarthy, Minsky, Rochester & Shannon's proposal

**1956**

**Dartmouth workshop**

The founding event of AI as a field

**1956**

**Logic Theorist**

Newell, Simon & Shaw: a program that proves theorems

# Read the proposal that named AI

Follow these steps:

1. Open the **Dartmouth proposal** (McCarthy et al., 1955) — it is only a few pages.
2. Find the sentence that first uses the words **‘artificial intelligence’**.
3. List the **seven problem areas** the authors planned to study.
4. Which of them are still unsolved today?

**Scan or click**



jmc.stanford.edu/…/dartmouth.pdf

**Open the link**

# The founders' generation



## John McCarthy

**1927–2011**

Coined ‘AI’; invented Lisp (1958), the classic AI language.



## Marvin Minsky

**1927–2016**

MIT AI Lab co-founder; frames for knowledge (1974).



## Herbert Simon

**1916–2001**

Logic Theorist & GPS; Nobel in economics (1978) for bounded rationality.



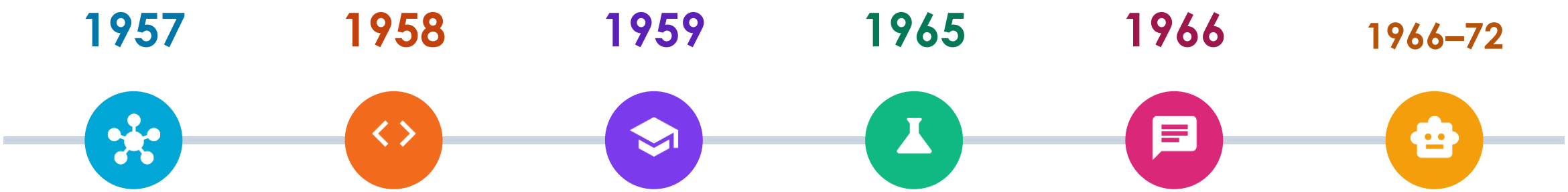
## Joseph Weizenbaum

**1923–2008**

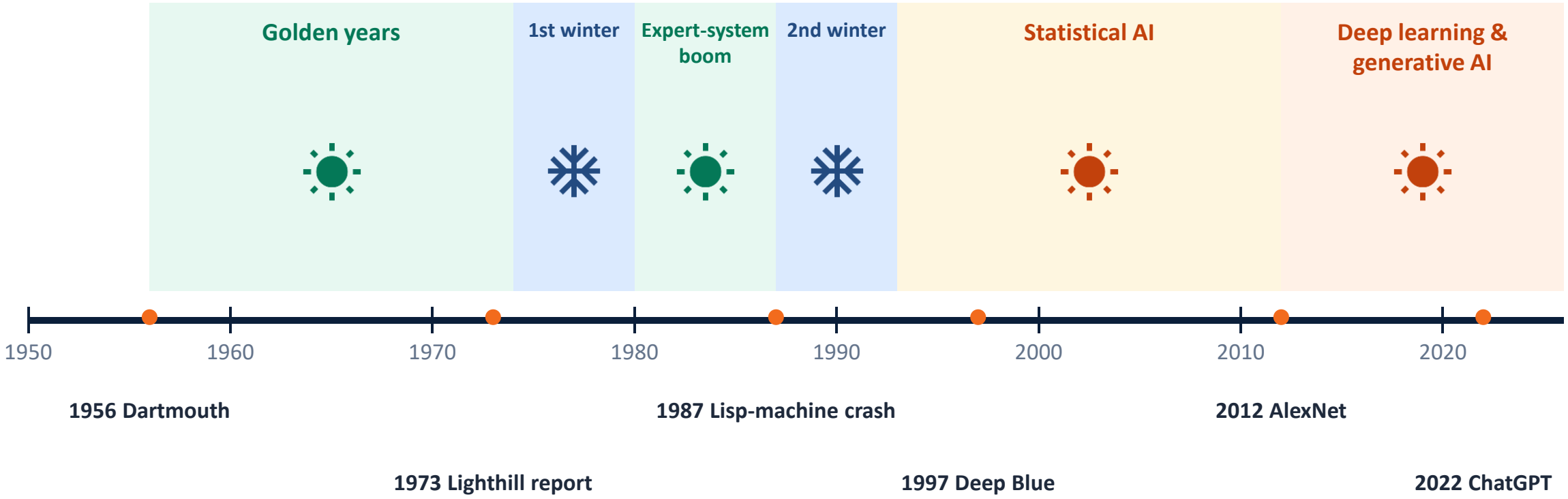
Wrote ELIZA (1966); later a strong critic of over-trusting computers.

*Photos: Wikimedia Commons (see the Image Credits slide for authors and licences).*

# 1956–1974: the golden years

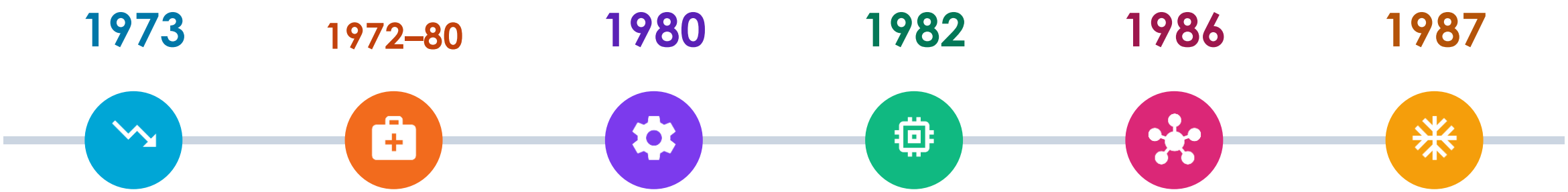


# AI summers and winters

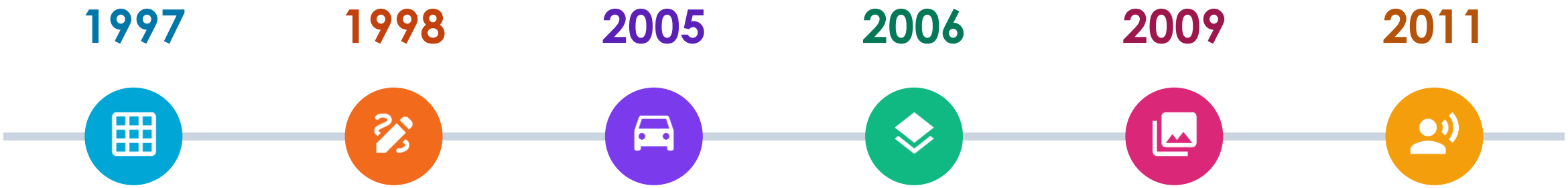



Lesson: promises outran results twice. Today's boom is driven by data, cheap computing and better algorithms — but over-hype is still a risk.

# 1974–1993: winters and the expert-system boom



# 1993–2011: statistical AI and big data



# Explore the history: Computer History Museum

Follow these steps:

1. Open the Computer History Museum's **‘Mastering the Game’** exhibit on computer chess.
2. Find the story of **Deep Blue** and of early chess programs.
3. Note one machine or person not mentioned in our slides.
4. Bonus: how did programs move from rules to learning?

**Scan or click**

computerhistory.org/chess

**Open the link**

# Faces of modern AI



## Garry Kasparov

**Chess champion**

Lost to Deep Blue (1997); now argues for human–AI teamwork.



## Fei-Fei Li

**Stanford**

Created ImageNet, the dataset that powered the deep-learning boom.



## Geoffrey Hinton

**Nobel Physics 2024**

Pioneer of neural networks and deep learning.



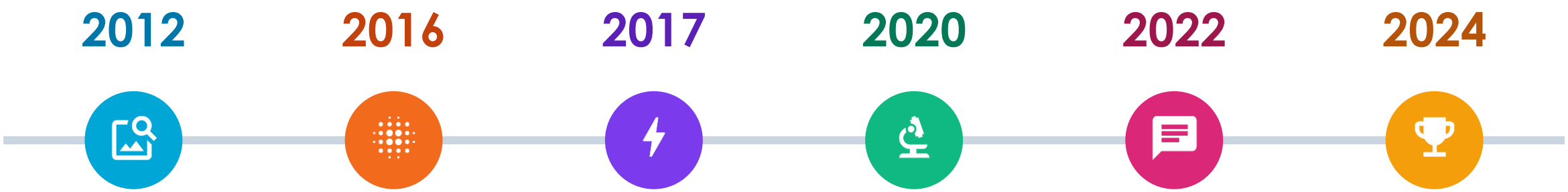
## Demis Hassabis

**Nobel Chemistry 2024**

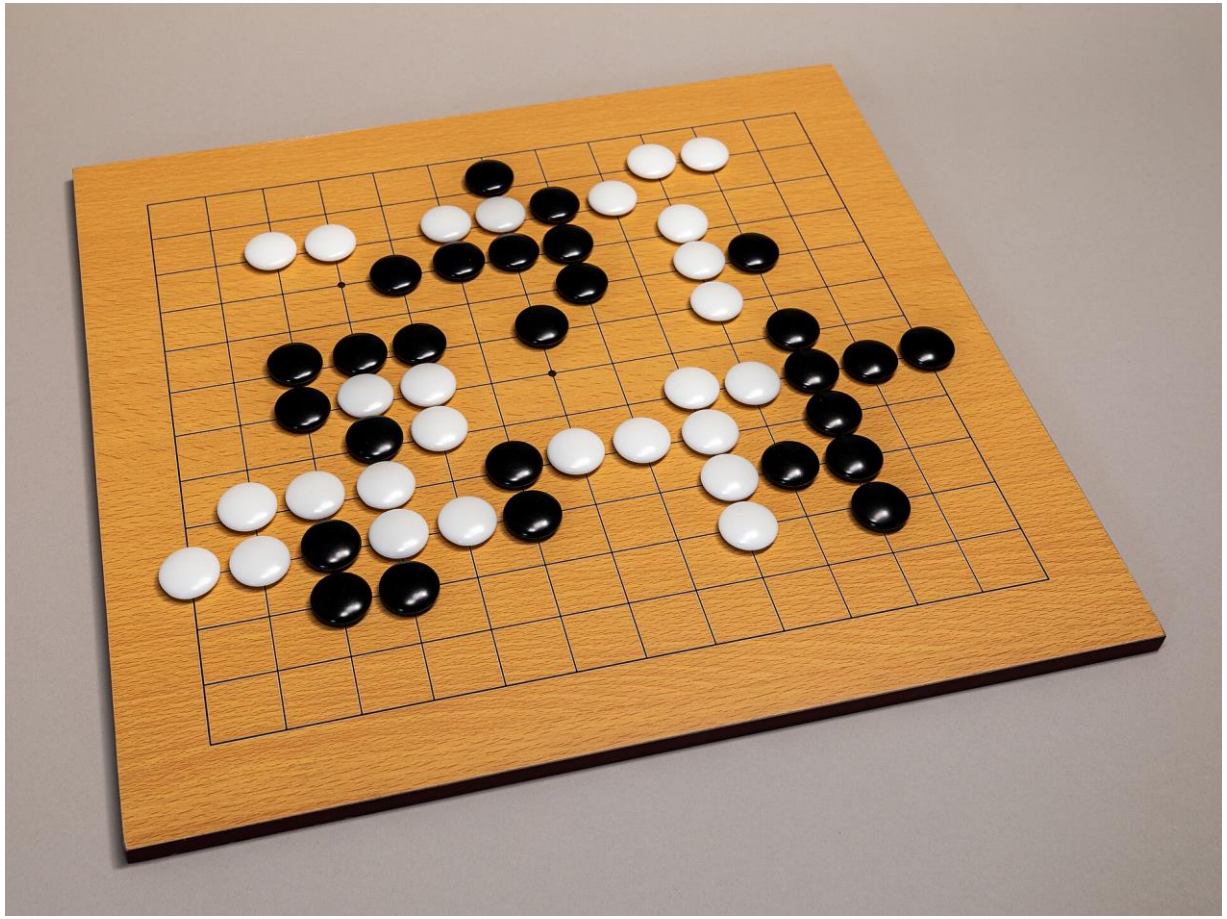
Co-founded DeepMind: AlphaGo and AlphaFold.

*Photos: Wikimedia Commons (see the Image Credits slide for authors and licences).*

# 2012–today: deep learning and generative AI



# AlphaGo (2016): beating humans at Go



*A game of Go (13 × 13 board)*

- Go has about **$10^{170}$** legal board positions — far more than chess.
- Brute-force search alone is hopeless (**combinatorial explosion**, Module 3).
- AlphaGo combined **deep neural networks** with **tree search** and learned from self-play.
- It beat **Lee Sedol 4–1** in Seoul, March 2016 — a decade earlier than many experts expected.

# Generative AI goes mainstream

**2017**

## Transformer

The neural-network design behind today's chatbots

**30 Nov 2022**

## ChatGPT released

A chatbot anyone could try in a browser

**~100 million**

## users in about 2 months

Estimated monthly users by January 2023

*Source: Vaswani et al. (2017); OpenAI; UBS estimate reported by Reuters (Feb 2023)*

# AI in Africa and Nigeria

**2016**



**Data Science Nigeria**

Free AI bootcamps; target: 1 million AI talents

**2017**



**Ubenwa**

Nigerian-founded: detecting birth asphyxia from a baby's cry

**2019**



**Google AI, Accra**

Africa's first Google AI research centre (Ghana)

**2020**



**NCAIR**

National Centre for AI and Robotics (NITDA), Abuja

**2024**



**National AI strategy**

Draft strategy; Nigerian multilingual LLM announced

**2024**



**N-ATLAS**

Open model for Yoruba, Hausa, Igbo & Nigerian English

# AI today: adoption is exploding

**78%**

## of organisations

used AI in 2024, up from 55% in 2023

**$109.1 billion**

## US private AI investment

in 2024 — about 12× China's $9.3 billion

**280×**

## cheaper

Cost of running a GPT-3.5-level model fell over 280× in 18 months

*Source: Stanford HAI, AI Index Report 2025*

# Goals of AI and the AI technique

